

Station geometry



DWG/DXF station overview

Evacuation demand

Reference occupants
+ train-arrival cohorts



INPUT

ARRIVAL-TIME-AWARE PLANNING AND EXECUTION

- capacity
- service rate
- receiving

- event time
- cohort size
- commitment

- current state
- queue state
- density

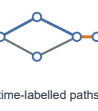
- route lock
- exit set
- update state

Dynamic flow state

ETA queue forecast

AA* path search

Rolling route assignment



$C_{uv}(\tau)$: movement + ETA queue + receiving + density exposure

DYNAMIC EVACUATION-FLOW EXECUTION

accepted routes $\rightarrow$ capacity competition $\rightarrow$ state update

EVACUATION PLAN

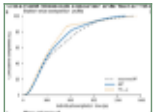
EXPORT

Route allocation



Pathfinder execution

Completion profile



cross-model check

OUTPUT